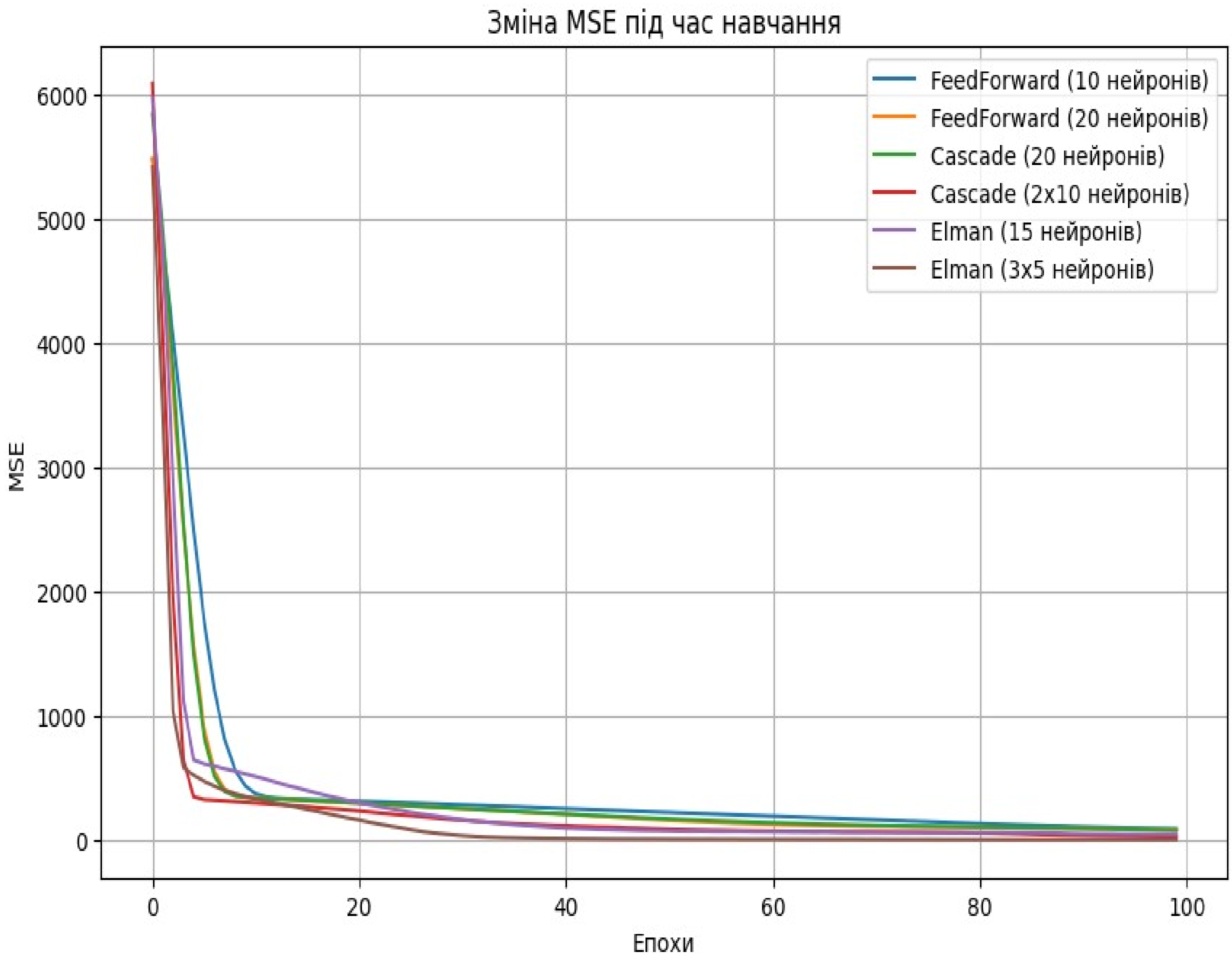
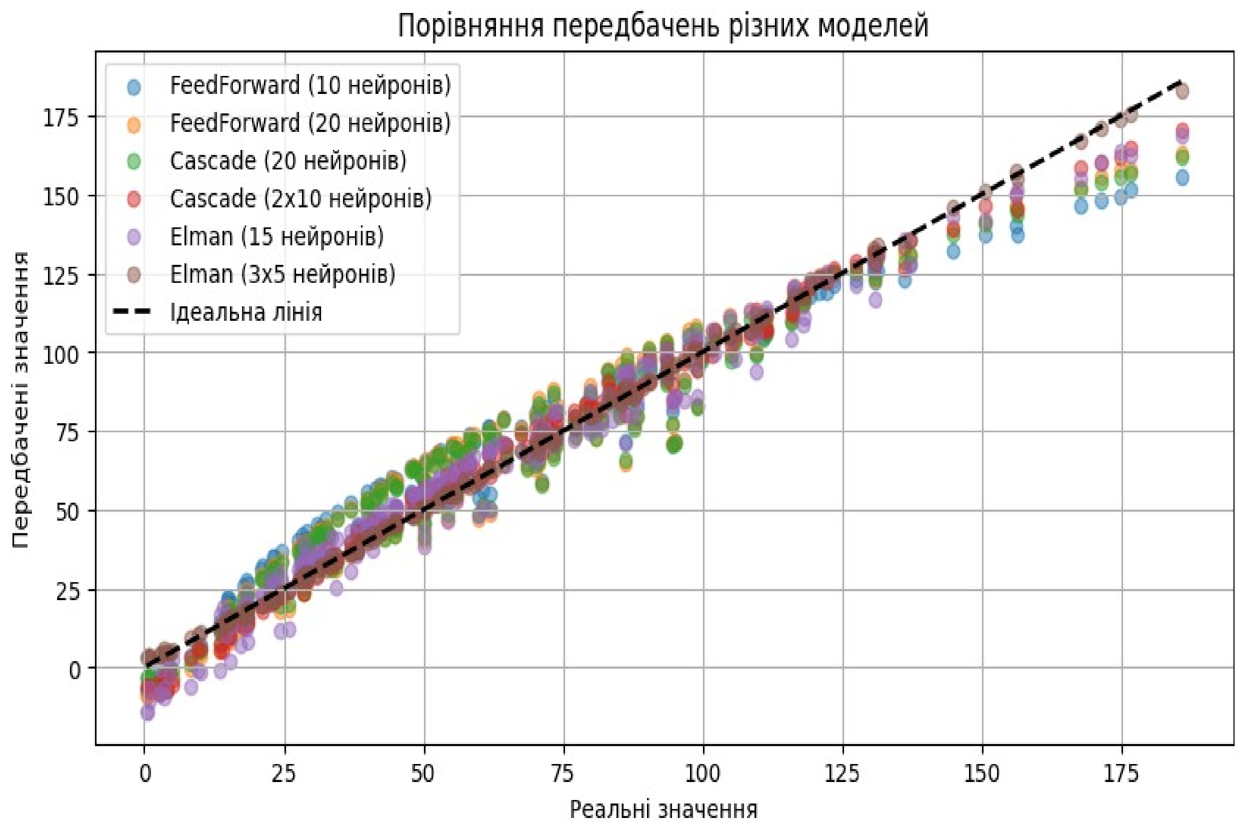
2026-05-21 21:11:37.365381: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1779397897.645341      57 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
E0000 00:00:1779397897.731293      57 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
W0000 00:00:1779397898.402383      57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779397898.402427      57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779397898.402430      57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1779397898.402433      57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
2026-05-21 21:12:08.452470: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
Навчання FeedForward (10 нейронів)...
Навчання FeedForward (20 нейронів)...
Навчання Cascade (20 нейронів)...
WARNING:tensorflow:5 out of the last 15 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f1b502327a0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
Навчання Cascade (2x10 нейронів)...
WARNING:tensorflow:5 out of the last 15 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7f1b396ccb80> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
Навчання Elman (15 нейронів)...
Навчання Elman (3x5 нейронів)...


Зміна MSE під час навчання
FeedForward (10 нейронів)
FeedForward (20 нейронів)
Cascade (20 нейронів)
Cascade (2x10 нейронів)
Elman (15 нейронів)
Elman (3x5 нейронів)
MSE
6000
5000
4000
3000
2000
1000
0
20
40
60
80
100
Епохи




=== Результати дослідження ===

FeedForward (10 нейронів): середня відносна помилка = 0.4946, $R^2$ = 0.9426

FeedForward (20 нейронів): середня відносна помилка = 0.5568, $R^2$ = 0.9469

Cascade (20 нейронів): середня відносна помилка = 0.3245, $R^2$ = 0.9531

Cascade (2x10 нейронів): середня відносна помилка = 0.3844, $R^2$ = 0.9865

Elman (15 нейронів): середня відносна помилка = 0.6137, $R^2$ = 0.9715

Elman (3x5 нейронів): середня відносна помилка = 0.1198, $R^2$ = 0.9989